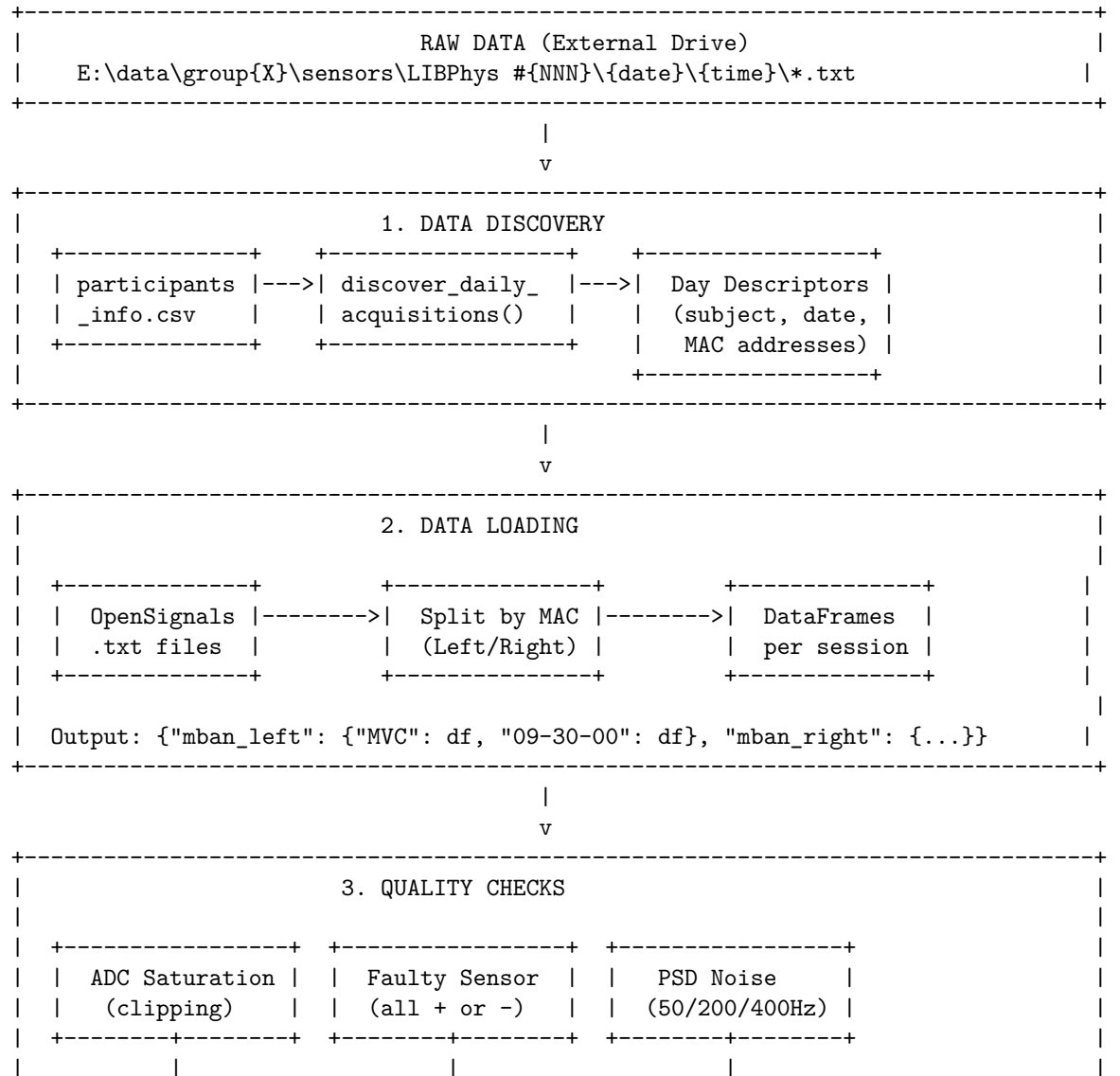


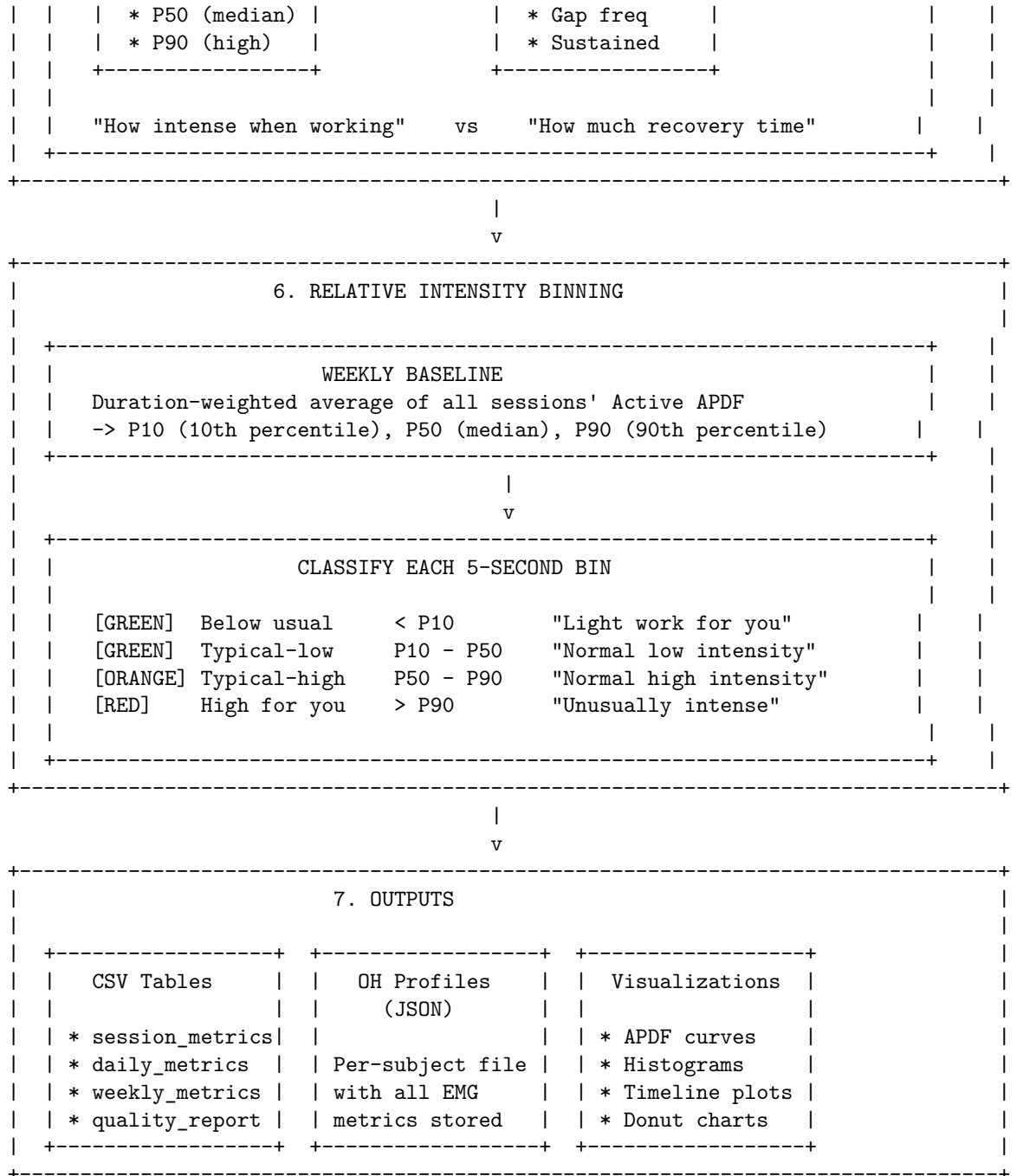
PrevOccupAI+ EMG Pipeline Overview

Purpose

Analyze surface EMG from trapezius muscles to assess occupational workload and generate personalized Occupational Health (OH) profiles.

High-Level Architecture





Code Organization

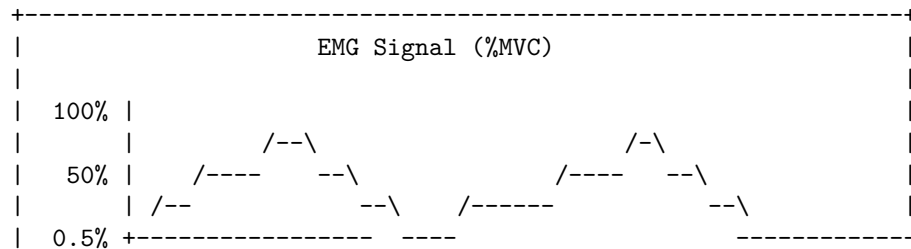
```

PrevOccupAI_Plus/
|
+-- main_emg.py          <-- ENTRY POINT
|
+-- sensors/             <-- ALL EMG CODE
|   |
|   +-- emg_pipeline.py  <-- Main orchestrator (run_emg_pipeline)
|   |
|   +-- load/            <-- Data Discovery & Loading
|       |
|       +-- dataset_loader.py * discover_daily_acquisitions()
|       |
|       +-- daily_data_loader.py * load_daily_acquisitions()
|       |
|       +-- path_handler.py * File path resolution
|       |
|       +-- data_quality.py * Quality report structures
|       |
|   +-- process/         <-- Signal Processing
|       |
|       +-- emg_preprocessing.py * Filtering, envelope, MVC RMS
|       |
|       +-- emg_mvc.py * MVC segment detection
|       |
|       +-- emg_quality_analysis.py * ADC saturation, PSD noise checks
|       |
|   +-- metrics/         <-- Metric Computation
|       |
|       +-- emg_metrics.py * APDF, rest metrics, aggregations
|       |
|       +-- emg_output.py * CSV export, table building
|       |
|   +-- visualize/       <-- Plotting
|       |
|       +-- emg_research.py * Signal-based plots (APDF, timeline)
|       |
|       +-- emg_oh.py * OH profile-based plots (donuts)
|
+-- OH_profile/          <-- JSON Profile Management
    +-- emg_oh_helper.py * Save EMG metrics to JSON
    +-- constants.py * Metric key definitions

```

Key Scientific Framework

Active APDF + Rest Time (Veiersted et al., 2013)



		+	---	ACTIVE	-----	+	+	-----	REST	-----	+	
				($\geq 0.5\%$ MVC)					($< 0.5\%$ MVC)			
				Compute APDF here					Count time here			
				(intensity when					(recovery/micro-breaks)			
				actually working)								
		+-----+-----+										

Key insight: Traditional APDF mixes “intensity when working” with “rest time”, making it hard to interpret. By separating them:

- **Active APDF** answers: “When this person is actually working, how hard?”
- **Rest metrics** answer: “How much recovery time are they getting?”

EMG Metrics (Detailed)

All EMG metrics are computed on %MVC signals (per session, per side). The pipeline uses a **rest threshold of 0.5% MVC** to separate active vs rest time. “Active” refers to samples $\geq 0.5\%$ MVC; “rest” refers to samples $< 0.5\%$ MVC.

1) Session Metadata (EMG_session)

- **duration_s**: Total recording duration in seconds.
- **mvc_peak**: MVC reference value used for normalization (peak RMS from MVC calibration).
- **active_duration_s**: Total time (seconds) with %MVC $\geq 0.5\%$.

2) Intensity Metrics (EMG_intensity)

- **mean_percent_mvc**: Mean %MVC across the whole session.
- **max_percent_mvc**: Maximum %MVC observed in the session.
- **min_percent_mvc**: Minimum %MVC observed in the session.
- **iemg_percent_seconds**: Integrated EMG in %MVC-seconds. Computed as:

$$\text{iEMG}_{\%MVC \cdot s} = \sum_{t=1}^N (\%MVC_t) \cdot \Delta t$$

3) APDF Percentiles (EMG_apdf)

APDF (Amplitude Probability Distribution Function) is computed for:

- **full**: includes all samples.

- **active**: includes only samples $\geq 0.5\%$ MVC.

For both **full** and **active**, the following percentiles are stored:

- **p10**: 10th percentile
- **p50**: 50th percentile (median)
- **p90**: 90th percentile

4) Rest & Recovery (EMG_rest_recovery)

- **rest_percent**: Percent of time below 0.5% MVC.
- **gap_frequency_per_minute**: Number of rest gaps per minute.
- **gap_count**: Total number of rest gaps.
- **max_sustained_activity_s**: Longest continuous active period (seconds).

Gap definition: contiguous rest segments with a minimum duration of 0.25 s.

5) Relative Intensity Bins (EMG_relative_bins)

Each 5-second bin is classified relative to the **weekly Active APDF baseline** (P10, P50, P90). Percentages are expressed over **active time only**:

- **below_usual_pct**: Active time $< P10$ (“light work for you”)
- **typical_low_pct**: Active time between $P10$ and $P50$
- **typical_high_pct**: Active time between $P50$ and $P90$
- **high_for_you_pct**: Active time $> P90$ (“unusually intense”)

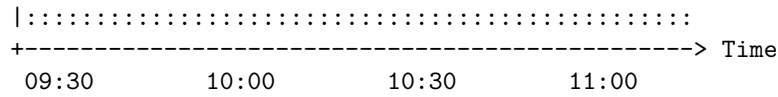
6) Aggregates

- **Daily aggregates** (EMG_daily_metrics): Same metric groups as above, summarized per day.
- **Weekly aggregates** (EMG_weekly_metrics): Same metric groups as above, summarized per week.
- **session_count**: number of sessions aggregated.
- **day_count**: number of days aggregated.

Example Output: Session Timeline

Subject 81 - Left Trapezius - 2024-09-30 09:30

```
EMG (%MVC)
|
50 |      ####                      #####
|      #####                      #####
25 | #####                      #####
| #####      #### #####
10 | #####
```



Legend:

High for you (>P90) :: Rest (<0.5% MVC)
 ## Typical-high (P50-P90) -- P50 baseline
 ## Typical-low (P10-P50) -- P10 baseline
 ## Below usual (<P10)

Clinical Relevance

Metric	What it tells us	Risk indicator
Active APDF P50	Typical muscle load when working	High = sustained strain
Rest %	Recovery time during work	Low = insufficient breaks
Gap frequency	Number of micro-pauses/min	Low = continuous tension
High-for-you %	Time above personal P90	High = overexertion risk

Running the Pipeline

```
# In main_emg.py
from main_emg import main

# Process all subjects
main(run_all=True)

# Process specific subjects
main(run_all=True, subject_filter=["81", "82", "83"])

# Quick test with limited data
main(run_all=True, max_subjects=2, max_days_per_subject=1)
```

References

- Veiersted, K. B., et al. (2013). “Assessment of occupational muscle activity.” *Applied Ergonomics*.

- Hansson, G. A., et al. (2010). “Validity and reliability of triaxial accelerometers for inclinometry in posture analysis.” *Medical and Biological Engineering and Computing*.

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